Question paper answer 2024

Part-1

2 marks

1. a) What is Data Mining?
Answer:
Data mining is the process of discovering meaningful patterns, trends, and relationships in large sets of data using statistical, machine learning, and database techniques. It helps in turning raw data into useful information for decision-making.
For example, businesses use data mining to analyze customer behavior and improve marketing strategies.

---

1. b) What is Supervised Learning and Unsupervised Learning?
Answer:

- **Supervised Learning** is a machine learning technique where the model is trained using labeled data. Example: Predicting house prices based on past data.
- **Unsupervised Learning** is used with unlabeled data to find hidden patterns.
  Example: Customer segmentation using clustering.
  Supervised is used for prediction, while unsupervised is used for pattern discovery.

---

1. c) Define Concept Hierarchy. Give an example.
Answer:
A concept hierarchy is a sequence of mappings from a set of low-level concepts to higher-level, more general concepts.
Example:
City → State → Country
(Mangalore → Karnataka → India)
It simplifies data analysis by allowing abstraction and summarization.

---

1. d) What is Data Mart? List its types.
Answer:
A data mart is a subset of a data warehouse focused on a specific business line or department, such as sales or finance. It provides faster access to relevant data.
Types of Data Marts:

1. Dependent Data Mart
2. Independent Data Mart
3. Hybrid Data Mart

1. e) What are Frequent Patterns? What is the use of Frequent Pattern Mining?
Answer:
Frequent patterns are itemsets or sequences that occur frequently in a dataset.
Example: In a supermarket, bread and butter often bought together is a frequent pattern.
Use:
Frequent pattern mining is used in market basket analysis, recommendation systems, and fraud detection.

---

1. f) What are Categorical Attributes and Quantitative Attributes?
Answer:

- **Categorical Attributes** represent categories or labels, like gender (male/female), or color (red, blue).
- **Quantitative Attributes** represent numeric, measurable values, like age, salary, or temperature.
  Categorical data is discrete, while quantitative data is continuous or discrete numerical.

---

1. g) What is Decision Tree?
Answer:
A decision tree is a tree-like model used for classification and prediction. It splits the dataset into branches based on attribute values, leading to decision outcomes.
Example:
If income > ₹50K → Approve loan
It is easy to understand and interpret, and widely used in machine learning.

---

1. h) Expand CLARA and ROCK.
Answer:

- **CLARA:** Clustering Large Applications
- **ROCK:** Robust Clustering using Links
  CLARA is used for clustering large datasets efficiently, while ROCK is a clustering algorithm for categorical data based on link connectivity.

PART-B

Unit 1

1. Define KDD Process. Explain the Different Stages of KDD

KDD (Knowledge Discovery in Databases) is the overall process of discovering useful knowledge from data. It involves selecting, preprocessing, transforming, and mining data to extract patterns, and then interpreting and evaluating them.

KDD is a broader concept than data mining. **Data mining is a step in the KDD process.**

Stages of KDD Process:

1. **Data Selection:**
   - Relevant data is selected from a larger database.
   - Only useful data for the mining task is chosen.
   - Example: Selecting customer transaction data from a retail database.
2. **Data Preprocessing / Cleaning:**
   - Removes noise, missing values, and inconsistencies from data.
   - Ensures data quality.
   - Example: Handling missing entries in a survey form.
3. **Data Transformation:**
   - Converts data into a suitable format for mining.
   - May involve normalization, aggregation, or encoding.
   - Example: Converting "Yes"/"No" into binary 1/0.
4. **Data Mining:**
   - Core step where intelligent methods are applied to extract patterns.
   - Techniques: Classification, Clustering, Association Rules, etc.
   - Example: Finding that customers who buy bread also buy butter.
5. **Pattern Evaluation:**
   - Filters out irrelevant or redundant patterns.
   - Only meaningful and interesting patterns are selected.
   - Example: Removing patterns that occur only once.
6. **Knowledge Presentation:**
   - Uses visualization, reports, or decision trees to present the discovered knowledge.

- Helps users understand and use the results effectively.

---

2. Explain Various Application Areas of Data Mining

Data mining is widely used in many fields to support decision-making and gain insights. Key application areas include:

1. **Retail and E-commerce:**
    - Market basket analysis (e.g., people who buy X also buy Y)
    - Customer segmentation
    - Recommendation systems (e.g., Amazon, Flipkart)
2. **Banking and Finance:**
    - Fraud detection
    - Credit risk analysis
    - Customer churn prediction
    - Loan approval prediction
3. **Healthcare and Medical Diagnosis:**
    - Predicting disease outbreaks
    - Diagnosing diseases using patient records
    - Discovering drug effectiveness patterns
4. **Education:**
    - Student performance prediction
    - Dropout analysis
    - Personalized learning paths
5. **Telecommunications:**
    - Customer retention strategies
    - Network fault diagnosis
    - Optimizing service quality
6. **Manufacturing and Production:**
    - Fault detection in machines
    - Quality control

  - Demand forecasting
7. **Social Media and Web Mining:**
   - Sentiment analysis
   - Trend detection
   - User behavior analysis
8. **Government and Crime Detection:**
   - Detecting tax fraud
   - Crime pattern analysis
   - National security monitoring

Here is a detailed yet exam-friendly answer for your long-answer question (suitable for 8–10 marks):

---

a) Explain the following Data Mining techniques:

---

i) Verification Model:

The **verification model** is a **user-driven** data mining approach where the user makes a **hypothesis** or assumption and uses data mining tools to **verify whether it is true or false**.

- It requires a prior understanding of the data or domain.
- The system only confirms or disproves the user's expectations.
- **Limitation:** It does not automatically discover new patterns.

Example:
A bank manager believes that customers aged 30–40 take more loans. The verification model checks this assumption by analyzing loan records.

---

ii) Discovery Model:

The **discovery model** is a **data-driven** approach where the system automatically **discovers hidden patterns or knowledge** from the data **without user assumptions**.

- It explores the data independently to find interesting trends or relationships.
- Often leads to new knowledge not previously known.

Example:
A retailer uses data mining and discovers that customers who buy diapers also often buy beer—an unexpected but useful association.

---

b) List and Explain the Issues and Challenges in Data Mining

Data mining faces several technical and practical challenges, including:

---

1. Data Quality and Integration

- Data may be **incomplete**, **noisy**, or **inconsistent**.
- Combining data from multiple sources (data integration) can be complex.
- Poor-quality data affects the accuracy of the results.

---

2. Scalability

- Modern databases are huge (terabytes or petabytes).
- Algorithms must be **efficient and scalable** to handle large datasets without excessive time or memory usage.

---

3. High Dimensionality

- Many datasets have hundreds or thousands of attributes (features).
- High dimensionality increases complexity and may lead to **overfitting** or **irrelevant patterns**.

---

4. Data Privacy and Security

- Mining personal or sensitive data raises ethical and legal concerns.
- Ensuring **privacy-preserving data mining** is a major challenge.

---

5. Real-Time Mining

- Some applications require mining to be done in **real time**, like stock trading or fraud detection.
- Ensuring fast, accurate results in real-time is difficult.

---

6. Interpretability of Results

- Complex patterns found by algorithms may be **hard to interpret**.
- Business users need **understandable and actionable** insights.

---

7. Domain Knowledge Dependency

- Effective mining often requires **domain-specific knowledge** to guide the process and interpret results meaningfully.

---

8. Dynamic Data and Evolving Patterns

- Data may **change over time**, requiring **updating models regularly**.
- Static models may become outdated and irrelevant.

UNIT-2

Here is a well-structured and clear answer for your **Question 4 (a, b, c)** — suitable for **8–10 marks** in an exam:

---

4. a) Define and Explain Data Warehouse

Definition:
A **Data Warehouse** is a subject-oriented, integrated, time-variant, and non-volatile collection of data that helps support decision-making processes. It stores historical data from multiple sources and is optimized for query and analysis, not transaction processing.

Explanation:

- **Subject-Oriented:** Organized around key subjects like sales, customers, or products.
- **Integrated:** Combines data from different sources into a unified format.
- **Time-Variant:** Maintains historical data (e.g., over months/years) for trend analysis.
- **Non-Volatile:** Data is stable and not frequently changed; only added through periodic loading.

Example:
A retail company may use a data warehouse to store and analyze sales data from various stores over several years to identify trends and improve marketing strategies.

---

4. b) Explain Any Four OLAP Operations

OLAP (Online Analytical Processing) provides tools for complex analytical queries. Four common OLAP operations are:

1. **Roll-up (Consolidation):**
   - Aggregates data by climbing up a concept hierarchy.
   - Example: Daily → Monthly → Yearly sales.
2. **Drill-down:**
   - Opposite of roll-up; moves from summary to detailed data.
   - Example: Yearly sales → Quarterly → Monthly.
3. **Slice:**
   - Selects a single dimension from a data cube, producing a sub-cube.
   - Example: Viewing sales data only for the year 2024.
4. **Dice:**
   - Selects data from multiple dimensions to form a sub-cube.
   - Example: Sales in Karnataka region for Q1 and Q2.

---

4. c) Explain Any Two Numerosity Reduction Techniques

Numerosity reduction techniques reduce the size of the data while preserving its integrity. Two common techniques are:

---

1. **Histograms:**
   - Divide data into bins or intervals and represent each bin with a summary value (like the average or frequency).
   - Example: Grouping ages 10–20, 21–30, etc., instead of storing each exact age.

---

2. **Clustering:**
   - Groups similar data points into clusters, and each cluster is represented by its centroid (mean) or representative value.
   - Helps in reducing the number of records while retaining important patterns.

Let me know if you'd like this in note/PDF format or want examples or diagrams added.